

the proper template for the future tense, the only thing they deign to do is change their single vowel to *a*. But these irregularities should not be dismissed too lightly, since there are once again good reasons to believe that the *a*-mutation is an extremely old pattern, a relic from the first steps that the ancestor of the Semitic languages was taking in developing the root-and-template system.

There are various clues within the Semitic languages themselves which suggest the extreme antiquity of the *a*-mutation, but perhaps the most compelling evidence comes from languages further afield. The Semitic languages are distantly related to some language families in Africa, including the Berber languages of Morocco and the Cushitic languages of Ethiopia and Somalia. Semitic, Berber and Cushitic are members of what scholars nowadays call the Afro-Asiatic language family. No one can say for sure when the Semitic branch of Afro-Asiatic started diverging from the Cushitic branch, but based on the linguistic distance between the languages, linguists believe that it must have been at the very least 8,000 years ago. While none of the other Afro-Asiatic languages has a root-and-template system like that of Semitic, many of them do show a suspiciously familiar vowel mutation between the tenses. In the Cushitic language Somali, for example, one comes across forms like these:

Somali (Cushitic, Afro-Asiatic)

<i>imid</i>	'I came'	<i>imādd</i>	'I (will) come'
<i>īl</i>	'I was in place'	<i>āl</i>	'I am (/will be) in place'
<i>iqān</i>	'I knew'	<i>aqān</i>	'I (will) know'

In fact, using the evidence from such verbs in various Cushitic languages, linguists have come to the conclusion that in the ancestor language of Cushitic (and perhaps of other Afro-Asiatic branches) there was a vowel mutation from *u* or *i* in the past tense to *a* in the present/future tense. So it would seem that the *a*-mutation goes back even beyond the earliest stratum of Semitic, to a time before Proto-Semitic started diverging from Proto-Cushitic, probably at least 8,000 years ago.

The evidence thus suggests that the *a*-mutation is an extremely old pattern. But if this is so, then why is this mutation only found in a few exceptional hollow verbs? Why is it that the regular verbs form their

future tense with an entirely different pattern (the template $a-(s)a(n)n(a)g$)? The most plausible explanation seems to be that the a -mutation was originally much more widespread, and was also the fashion among the normal three-consonant verbs. So, for example, the future tense of the verb $a-ktum$ 'I covered' would simply have been $a-ktam$. But at a later stage, the a -mutation was displaced by the more elaborate future template (as a part of the general overhaul the system underwent with the development of the mature root-and-template system). The few Akkadian verbs that still show the a -mutation are just the last survivors, those which managed to hold out against the new template most obstinately. So even if from the later perspective of the mature Semitic system the a -mutation may look like nothing more than an unmotivated and rather embarrassing irregularity, the a -mutation was probably around long before the other templates had even been thought of.

Let's now put together everything we have uncovered so far: some time in the prehistoric period, the ancestor of the Semitic languages must have started out with 'normal' verbs, with sturdy stems like $mūt$, $nīk$, $ktum$ or $ptil$, which had both consonants and vowels to their name. The first step in the evolution of the root-and-template design may have been taken as early as 8,000 years ago, when, for some strange reason, a vowel mutation emerged in the future tense: the vowel of the ancient stem changed to a :

ANCIENT STEM	SIMPLE PAST	SIMPLE FUTURE
$ktum$ ('cover')	$a-ktum$ ('I covered')	$a-ktam$ ('I will cover')
$mūt$ ('die')	$a-mūt$ ('I died')	$a-māt$ ('I will die')
$ptil$ ('twist')	$a-ptil$ ('I twisted')	$a-ptal$ ('I will twist')

In itself, this mutation pattern may not seem such a huge leap forward. Nevertheless, the a -mutation is a defining moment in the evolution of the Semitic verbal design, since the kernel of a new concept has been formed, from which the notion of the consonantal root will later spring: the idea that a verb can keep the same consonants, but change the vowels between them to mark nuances like tense.

Now this is all very well, of course, but suggesting that the first step was the emergence of one vowel mutation still doesn't say anything about *how* this first step could ever have been taken. What could have

galvanized the change of vowel in the future tense from *i* or *u* to *a*? The next clue will help us tackle exactly this question.

CLUE 3: GEESE, GUESTS AND GERMAN CARDINALS

The clues so far have relied mostly on evidence from within the Semitic language family, but in order to understand how the *a*-mutation could have developed, the best clues actually come from parallels much closer to home. Are there any familiar languages which have a pattern similar to the *a*-mutation? The first example that springs to mind, of course, is those English verbs like *sit-sat*, *drink-drank*, or German ones like *trinken-tranken*. Unfortunately, however, the origin of the vowel mutation in the Germanic verbs is also pretty obscure, as it goes back to the deepest strata of Proto-Indo-European. But we shouldn't give up on Germanic just yet, because there happens to be another pattern of vowel mutation in the Germanic languages, whose origin is more recent and thus better understood. Consider the following pairs:

ENGLISH	GERMAN	
<i>goose – geese</i>	<i>Gast – Gäste</i>	('guest – guests')
<i>man – men</i>	<i>Hals – Hälse</i>	('neck – necks')
<i>tooth – teeth</i>	<i>Nacht – Nächte</i>	('night – nights')
<i>foot – feet</i>	<i>Fuss – Füße</i>	('foot – feet')

At first, messing around with a bunch of badly behaved Germanic nouns may seem rather a detour from the quest for the origins of the Semitic verb. Nevertheless, I promise that wandering through the thickets of Germanic philology will soon bring us to where we want to be. For whether in Germanic or Semitic, whether in nouns or verbs, our goal is to understand how a vowel mutation with a *grammatical* function can emerge. And once we have worked out how one vowel mutation has come to mark one grammatical function in one family, it will become much easier to grasp how another vowel mutation could have come to mark a different grammatical function in another family.

To see how the vowel mutation developed in the Germanic nouns, it is easier to start off with German rather than with English. Comparative

evidence suggests that initially, there was nothing unusual about the noun *gast* ('guest'), and that its plural was simply **gast-iz* (*-iz* was a normal plural ending of Proto-Germanic). But at some stage before the earliest records of German in the eighth century AD, effort-saving mechanisms were set in motion and a few sound changes occurred. One was a type of assimilation (the 'Santa Siesta' principle from Chapter 3). In the word *gastiz*, there were two different vowels in close proximity, which required two very different configurations of the mouth. Moving the tongue quickly from a shape needed for an *a* {ah} to an *i* {ee} is really quite a bother, so to save effort, the first vowel *a* was 'coloured' by the *i*, and changed into something half-way between, namely to *e* {eh}. So *gastiz* became *gestiz*. Linguists call this process *i*-mutation, because the *i* caused the preceding vowel to change.

The final *z* of *gestiz* was also dropped at some stage, to give the plural form *gesti*, which is what we find in the first records of German in the eighth century. But the effort-saving spree did not come to an end right there. In the eleventh century, the final *-i* of *gesti* was weakened further to just a *schwa*, a reduced indistinct vowel (written *ə* in phonetic transcription) which can be heard in English words like *elephant* {eləfənt} or 'bother' {boθə}. And so *gesti* ended up as *gestə*, giving the pair *gast-gestə*. (In modern German orthography, *gestə* is spelled *Gäste*, but for simplicity, a phonetic spelling will be used here instead.) In other words, after the ending *iz* messed things up by changing *gastiz* to *gestiz*, this ending itself fell victim to the forces of erosion and disappeared, leaving behind a mere schwa.

So far, these two changes were entirely mechanical effort-saving devices. They had nothing to do with the *meaning* of the plural – in fact, they had nothing to do with any kind of meaning whatsoever. They were blind changes, influenced only by the phonetic environment. Nevertheless, the combination of these blind changes created the potential for a new meaningful pattern to emerge. The plural form *gestə* was stuck with an *e* in the middle (instead of the original *a*), but there was no longer anything to remind speakers of why this *e* had come to be there in the first place. The order-craving minds of a new generation of speakers could thus seize on the pattern *gast-gestə*, and assume that the change of vowel from *a* to *e* must be there for some purpose, and that purpose must be to indicate plurality.

And once the mutation from *a* to *e* was perceived to be a meaningful pattern, speakers could also extend it by analogy to other nouns, even those which from a historical perspective were not likely candidates. For example, the noun *hals* ('neck') originally had a different plural ending *-az*, so its plural form was originally **halsaz*, not *halsiz* – no reason for *i*-mutation there. If it were only down to effort-saving changes, then, the plural *halsaz* should have ended up as *halsə*. But the new generation of German speakers couldn't give a sausage about the 'historical perspective', so on the analogy of nouns like *gast-gestə*, they coined the pair *hals-helsə*:



Even newly borrowed nouns were subjected to this analogy. For example, the nouns *kardinal* and *general* entered German only in the thirteenth century, long after the original culprit for *i*-mutation (the ending *-iz*) had disappeared. But if the plural of *gast* is *gestə*, and the plural of *hals* has now come to be *helsə*, then what should be the plural of the newcomers *kardinal* and *general*, if not *kardinelə*, and *generelə*? (In modern orthography, these are spelled *Kardinäle* and *Generäle*.)

To summarize, then, what created the pattern of vowel mutation (*a* → *e*) as a marker of plurality was a cycle of erosion and analogy. A sequence of effort-saving changes created the conditions for the appearance of this pattern (first, the ending *-iz* coloured the previous vowel *a* to *e*, to give *gestiz*, and then this ending itself was eroded out of all recognition, to leave just *gestə*). But it was not erosion on its own that turned these blind changes into a *meaningful* distinction. The grammatical function of the pattern was the brain-child of the order-craving mind.

In English, incidentally, the development initially ran along similar lines, except that it started a few centuries earlier than German. The noun *man* (originally *mann*) must have started with a regular plural form **mann-iz*, but then the effects of the vowel *i* changed the preceding *a* to